

1. Tên học phần/Course name: **Statistics & Probability_Xác suất thống kê**

2. Thời gian triển khai: Học kỳ: **Summer 2025**

3. Lịch triển khai môn học:

Week	Topic	Session	nSlot	Content	Traditional/SC	Mark
01: 12/05 - 18/05	1. The Roles of Statistics in Engineering	1	1	The Roles of Statistics in Engineering 1.1 The Engineering Method and Statistical Thinking 1.2 Collecting Engineering Data	Traditional	
		2		1.3 Mechanistic and Empirical Models 1.4 Probability and Probability Models		
		3		Probability 2.1 Sample Spaces and Events		
	2. Probability	4	2	Probability 2.2 Interpretations of Probability 2.3 Addition Rules	SC	
		5		2.4 Conditional Probability		
		6		2.5 Multiplication and Total Probability Rules 2.6 Independence		
02: 19/05 - 25/05	2. Probability	7	3	2.7 Bayes' Theorem	SC	
		8		Probability 2.8 Random Variables		
		9		Discrete Random Variables and Probability Distribution 3.1 Discrete Random Variables		
	3. Discrete Random Variables and Probability Distribution	10	4	3.2 Probability Distributions and Probability Mass Functions	Traditional	A1
		11		3.3 Cumulative Distribution Functions		
		12		3.4 Mean and Variance of a Discrete Random Variable		
03: 26/05 - 01/06	3. Discrete Random Variables and Probability Distribution	13	5	3.5 Discrete Uniform Distribution 3.6 Binomial Distribution	SC	
		14		3.7 Geometric and Negative Binomial Distributions		
		15		3.8 Hypergeometric Distribution		
		16	6	3.9 Poisson Distribution	SC	
		17		Solve questions and problems for Chapters 1 - 3		
		18		Progress Test 1		A2
04: 02/06 - 08/06	4. Continuous Random Variables and Probability Distribution	19	7	Continuous Random Variables and Probability Distribution 4.1 Continuous Random Variables 4.2 Probability Distributions and Probability Density Functions	SC	
		20		4.3 Cumulative Distribution Functions		
		21		4.4 Mean and Variance of a Continuous Random Variable		
		22	8	4.5 Continuous Uniform Distribution	Traditional	
		23		4.6 Normal Distribution		
		24		4.6 Normal Distribution (Cont.)		

05: 09/06 - 15/06	4. Continuous Random Variables and Probability Distribution	25	9	4.7 Normal Approximation to the Binomial and Poisson Distributions	SC	
		26		4.8 Exponential Distribution		
		27		Descriptive Statistics		
	6. Descriptive Statistics	28	10	6.2 Stem-and-Leaf Diagrams	SC	T1
		29		6.3 Frequency Distributions and Histograms		
		30		6.4 Box Plots		
06: 16/06 - 22/06	7. Sampling Distributions and Point Estimation of Paramaters	31	11	6.5 Time Sequence Plots	SC	
		32		Sampling Distributions and Point Estimation of Paramaters		
		33		7.1 Point Estimation		
	8. Statistical Intervals for a Single Sample	34	12	7.2 Sampling Distributions and the Central Limit Theorem	SC	T2
		35		Solve questions and problems for Chapters 4, 6, and 7		
		36		Progress Test 2		
07: 23/06 - 29/06	8. Statistical Intervals for a Single Sample	37	13	Statistical Intervals for a Single Sample	SC	
		38		8.1 Confidence Interval on the Mean of a Normal Distribution, Variance Known		
		39		Statistical Intervals for a Single Sample		
	9. Test of Hypotheses for a Single Sample	40	14	8.1 Confidence Interval on the Mean of a Normal Distribution, Variance Known (Cont.)	Traditional	T2
		41		8.2 Confidence Interval on the Mean of a Normal Distribution, Variance Unknown		
		42		8.2 Confidence Interval on the Mean of a Normal Distribution, Variance Unknown (Cont.)		
08: 30/06 - 06/07	9. Test of Hypotheses for a Single Sample	43	15	8.4 Large-Sample Confidence Interval for a Population Proportion	SC	
		44		8.4 Large-Sample Confidence Interval for a Population Proportion (Cont.)		
		45		Test of Hypotheses for a Single Sample		
	10. Statistical Inference for Two Samples	46	16	9.1 Hypothesis Testing	SC	
		47		9.1 Hypothesis Testing (Cont.)		
		48		9.2 Tests on the Mean of a Normal Distribution, Variance Known		
09: 14/07 - 20/07	10. Statistical Inference for Two Samples	49	17	9.2 Tests on the Mean of a Normal Distribution, Variance Known (Cont.)	SC	T3
		50		9.3 Tests on the Mean of a Normal Distribution, Variance Unknown		
	11. Simple Linear Regression and	51		9.5 Tests on a Population Proportion		
				Statistical Inference for Two Samples		
				10.1 Inference on the Difference in Means of Two Normal Distributions, Variance Known		
				10.2 Inference on the Difference in Means of Two Normal Distributions, Variance Unknown		
				10.6 Inference on the Two Proportions		
				10.6 Inference on the Two Proportions (Cont.)		
				Simple Linear Regression and Correlation		
				11.1 Empirical Models		

	Correlation					
	11. Simple Linear Regression and Correlation	52	18	11.2 Simple Linear Regression	sc	
		53		11.3 Properties of the Least Squares Estimators		
		54		11.4 Hypothesis Tests in Simple Linear Regression		
10: 21/07 - 27/07	11. Simple Linear Regression and Correlation	55	19	11.8 Correlation	Traditional	CP
		56		11.8 Correlation (Cont.)		
		57		Solve questions and problems for Chapters 8 - 11		
	Revision and Wrap-up	58	20	Progress Test 3	sc	
		59		Project presentation		
		60		Revision and Wrap-up		
	Final test (60 minutes, 50 questions)					
Assignment(20%): A1 10%, A2 10% , Computer Project: 15%, Progress Test(30%): T1 10%, T2 10%, T3 10%, Final Example: 35%						

Người phê duyệt/Approver GĐCS/Campus's Director	Người kiểm tra/Reviewer TBĐT/Head of Academic Affairs Board)	Người lập/Creator
Họ tên/Name: Trần Ngọc Anh Ngày/Date: 05/2025	Họ tên/Name: Ngày/Date:	Họ tên/Name: Huỳnh Tiến Sĩ Ngày/Date: 01/05/2025